True Delta-zero sentences are provable in Q

The Open Logic Project contributors

openlogic:incompleteness/representability-in-q:local-snapshot-20261013; CC BY 4.0

Editorial note. The counted statement is the author lemma labelled lem:delta0-completeness (2058-2062); the proof span 2064-2133 is the author proof environment that immediately follows it.

Editorial note. External reference numbers are set to the source labels themselves so the delivered card shows the exact unit-internal pointer instead of an invented display number.

Editorial note. External reference numbers render the unit-internal labels with hyphens in place of colons (pipeline number charset); the Source reference lines below the body give each label's exact source span.

Editorial scope. One original result of the Open Logic Project unit incompleteness/representability-in-q, counted once: true Delta_0 sentences are provable in Q. Only the counted lemma and its own author proof are retained; the source is reproduced without mathematical changes and keeps the CC BY 4.0 licence and the author attribution.

Allowed standard prerequisites (editorial).

- Delta_0 sentences. A Delta_0 sentence is built from atomic formulas using propositional connectives and quantifiers that are all bounded by closed terms.. The induction ranges over exactly this class of sentences, and each enumerated case matches one constructor.
- Induction on formula complexity. Each case may assume the counted statement for all immediate subformulas of the sentence under consideration.. Every case except the base invokes the induction hypothesis on subformulas (2064-2133).
- Atomic completeness in Q. For a true closed atomic sentence, Q proves it; for a false one, Q proves its negation; the unit establishes this as its own lemma.. It supplies the base case (2069-2070) and the atomic subcase of the final negation case (2124-2127).
- Bounded quantifiers versus finite instances. Q proves that a bounded universal quantifier over a closed-term bound is equivalent to the finite conjunction of its numeral instances, and dually for the bounded existential; the unit establishes this as its own lemma.. It converts the bounded-quantifier cases (2096-2117) into finite conjunctions and disjunctions already handled by the induction hypothesis.

Source reference. Original lemma lem-atomic-completeness: sources/olp-incompleteness-representability-in-q/pinned-original-source.tex, source lines 1955–1955. The author proof cites the unit's atomic completeness lemma for the base case and for the atomic subcase of the final negation case. The load-bearing work is carried by that separately proved lemma of the same unit, which is itself staged as a card; the counted proof retains its own nine-case argument in full.

Source reference. Original lemma lem-bounded-quant-equiv: sources/olp-incompleteness-representability-in-q/pinned-original-source.tex, source lines 2011–2011. The author proof cites the unit's bounded-quantifier equivalence lemma to pass between a bounded quantifier and its finite numeral instances. The load-bearing work is carried by that separately proved lemma of the same unit; the counted proof retains its own case analysis in full.

Lemma 0.1. *If* $!A$ *is a* Δ_0 *!!sentence which is true in* $\mathfrak{A}N$*, then* $\mathrm{Th}Q \vdash !A$*.*

Proof. We prove this by induction on !!formula complexity. The base case is given by lem-atomic-completeness, so we move to the induction step. For simplicity we split the case of negation into subcases depending on the structure of the !!formula to which the negation is applied.

1. Suppose $(!A \wedge !B)$ is true in $\mathfrak{A}N$, so $!A$ and $!B$ are true in $\mathfrak{A}N$. By the induction hypothesis, $\mathrm{Th}Q \vdash !A$ and $\mathrm{Th}Q \vdash !B$, so $\mathrm{Th}Q \vdash (!A \wedge !B)$ by logic.

2. Suppose $\neg(!A \wedge !B)$ is true in $\mathfrak{A}N$, so either $\neg !A$ or $\neg !B$ is true in $\mathfrak{A}N$. Without loss of generality, suppose the former. By the induction hypothesis $\mathrm{Th}Q \vdash \neg !A$, and hence $\mathrm{Th}Q \vdash \neg(!A \wedge !B)$ by logic.

3. Suppose $(!A \vee !B)$ is true in $\mathfrak{A}N$, so either $!A$ is true in $\mathfrak{A}N$ or $!B$ is true in $\mathfrak{A}N$. Without loss of generality, suppose the former holds. By the induction hypothesis $\mathrm{Th}Q \vdash !A$, and hence $\mathrm{Th}Q \vdash (!A \vee !B)$ by logic.

4. Suppose $\neg(!A \vee !B)$ is true in $\mathfrak{A}N$, so $\neg !A$ and $\neg !B$ are true in $\mathfrak{A}N$. Then $\mathrm{Th}Q \vdash \neg !A$ and $\mathrm{Th}Q \vdash \neg !B$ by the induction hypothesis. Consequently, $\mathrm{Th}Q \vdash \neg(!A \vee !B)$ by logic.

5. Suppose that $\forall x < t !A(x)$ is true in $\mathfrak{A}N$, where t is a closed term and $k = \mathrm{val}(t)N$. By the induction hypothesis and logic, if $!A(\underline{n})$ is true in $\mathfrak{A}N$ for all $n < \mathrm{val}(t)N$ then $\mathrm{Th}Q \vdash !A(\underline{0}) \wedge \cdots \wedge !A(\underline{k-1})$. By lem-bounded-quant-equiv it follows that $\mathrm{Th}Q \vdash \forall x < t !A(x)$.

6. The case for the bounded existential quantifier, where we have !!asentence of the form $\exists x < t !A(x)$, is similar to that for the bounded universal quantifier.

7. Suppose that $\neg\forall x < t !A(x)$ is true in $\mathfrak{A}N$, where t is a closed term. This !!sentence is equivalent to the !!sentence $\exists x < t \neg !A(x)$, with the equivalence derivable in $\mathrm{Th}Q$, so we may apply the reasoning for bounded existential quantifiers.

8. Similarly, suppose that $\neg\exists x < t !A(x)$ is true in $\mathfrak{A}N$, where t is a closed term. This !!sentence is equivalent in $\mathrm{Th}Q$ to $\forall x < t \neg !A(x)$, and so we may apply the reasoning for bounded universal quantifiers.

9. Finally, suppose $\neg!A$ is true in $\mathfrak{A}N$. The only cases remaining are when $!A$ is atomic and when $\neg!A \operatorname{id} \neg\neg!B$ for some Δ_0 !!sentence $!B$. If $!A$ is atomic then by lem-atomic-completeness, $\mathrm{Th}Q \vdash \neg!A$. If $\neg!A \operatorname{id} \neg\neg!B$, then by logic it is provably equivalent in $\mathrm{Th}Q$ to $!B$, which is true in $\mathfrak{A}N$ since $\neg!A$ is true in $\mathfrak{A}N$. By the induction hypothesis we therefore have that $\mathrm{Th}Q \vdash \neg!A$.

□

Proof. We prove this by induction on !!formula complexity. The base case is given by lem-atomic-completeness, so we move to the induction step. For simplicity we split the case of negation into subcases depending on the structure of the !!formula to which the negation is applied.

1. Suppose $(!A \wedge !B)$ is true in $\mathfrak{A}N$, so $!A$ and $!B$ are true in $\mathfrak{A}N$. By the induction hypothesis, $\mathrm{Th}Q \vdash !A$ and $\mathrm{Th}Q \vdash !B$, so $\mathrm{Th}Q \vdash (!A \wedge !B)$ by logic.

2. Suppose $\neg(!A \wedge !B)$ is true in $\mathfrak{A}N$, so either $\neg!A$ or $\neg!B$ is true in $\mathfrak{A}N$. Without loss of generality, suppose the former. By the induction hypothesis $\mathrm{Th}Q \vdash \neg!A$, and hence $\mathrm{Th}Q \vdash \neg(!A \wedge !B)$ by logic.

3. Suppose $(!A \vee !B)$ is true in $\mathfrak{A}N$, so either $!A$ is true in $\mathfrak{A}N$ or $!B$ is true in $\mathfrak{A}N$. Without loss of generality, suppose the former holds. By the induction hypothesis $\mathrm{Th}Q \vdash !A$, and hence $\mathrm{Th}Q \vdash (!A \vee !B)$ by logic.

4. Suppose $\neg(!A \vee !B)$ is true in $\mathfrak{A}N$, so $\neg!A$ and $\neg!B$ are true in $\mathfrak{A}N$. Then $\mathrm{Th}Q \vdash \neg!A$ and $\mathrm{Th}Q \vdash \neg!B$ by the induction hypothesis. Consequently, $\mathrm{Th}Q \vdash \neg(!A \vee !B)$ by logic.

5. Suppose that $\forall x < t!A(x)$ is true in $\mathfrak{A}N$, where t is a closed term and $k = \mathrm{val}(t)N$. By the induction hypothesis and logic, if $!A(\underline{n})$ is true in $\mathfrak{A}N$ for all $n < \mathrm{val}(t)N$ then $\mathrm{Th}Q \vdash !A(\underline{0}) \wedge \cdots \wedge !A(\underline{k-1})$. By lem-bounded-quant-equiv it follows that $\mathrm{Th}Q \vdash \forall x < t!A(x)$.

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9. Finally, suppose $\neg!A$ is true in $\mathfrak{A}N$. The only cases remaining are when $!A$ is atomic and when $\neg!A \operatorname{id} \neg\neg!B$ for some Δ_0 !!sentence $!B$. If $!A$ is atomic then by lem-atomic-completeness, $\mathrm{Th}Q \vdash \neg!A$. If $\neg!A \operatorname{id} \neg\neg!B$, then by logic it is provably equivalent in $\mathrm{Th}Q$ to $!B$, which is true in $\mathfrak{A}N$ since $\neg!A$ is true in $\mathfrak{A}N$. By the induction hypothesis we therefore have that $\mathrm{Th}Q \vdash \neg!A$.

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